

Product Requirement Document (PRD) — howfar.today

1. Project Vision & Architectural Philosophy

howfar.today is a hyper-minimalist, single-page, full-screen day-virtualizer designed to balance daily planning with real-time progress awareness. The application embodies contemporary flat minimalism—deliberately rejecting shadows, rounded corners, complex borders, or non-functional visual noise.

Core Architectural Principle: Extensible & Scalable Foundation

- **Not a Quick Single-Use App:** Although launching as a highly polished, single-file MVP, the codebase must be structured as a decoupled, robust framework. All visual rules, time calculations, state engines, and text layouts should remain highly modular.
- **Separation of Concerns:** Business logic, timing systems, rendering calculations, and configuration models must be cleanly separated to allow seamless integration of future capabilities—such as cloud database synchronization, advanced calendar integrations, dragging/splitting tasks, multi-user real-time views, and server-side features.

2. Page & Layout Architecture

- **Layout Structure:** A single-page, full-bleed viewport split into exactly three stacked slides. Each slide occupies precisely $100vw$ by $100vh$ ($100dvw$ / $100dvh$ where supported to prevent mobile browser chrome shifts).
- **Relative Sizing Mandate:** All dimensions, margins, paddings, strokes, and font sizes are defined in fluid relative units (vw , vh , $\%$) to guarantee flawless, pixel-perfect visual rendering regardless of desktop or mobile browser sizing.
- **Full-Page Slide Navigation:**
 - **Scroll-Snapping:** Scrolling (mouse wheel, touch swipe, or trackpad) triggers immediate, snap-locked navigation to the preceding or succeeding slide. Transitions use a clean, modern, hardware-accelerated ease-out animation.
 - **System-Level Page Keyboards:** Up and Down arrow keys transition sequentially between slides when the user is not actively editing a grid input field.

3. Screen Specifications

Slide 1: Focus Mode (Top Screen)

Designed for distraction-free execution of the active time slot.

- **Dynamic Full-Screen Progress Fill:** The absolute background of the slide acts as a horizontal progress bar. It fills smoothly from left (0%) to right (100%) based on the current segment's elapsed seconds.

- **Header Metadata:** Displays a clean, prominent text indicating the exact bounds of the running segment (e.g., 07:15 – 07:30).
- **Centered Goal Display:** Centered, highly visible large typography displaying the goal/task defined for the active segment.
- **Interactive Bottom Status Button:** A minimal checkmark button positioned at the bottom center.
 - Click/tap transitions the segment state immediately to "Completed" and transforms the button into a "revert" button.
 - Completion updates the full-bleed progress fill color instantly to the configured Success color token.
 - If the segment's duration elapses and the user has not marked it as completed, the segment permanently locks as "Failed" (if the status coloring is toggled active).

Slide 2: Day Visualizer (Middle / Main Screen)

The visual centerpiece of the web application, mapping out the entire configured day.

- **Grid Architecture:** The dynamic layout divides the vertical viewport height (100vh) into equal horizontal rows based on user parameters (e.g., from 07:00 to 24:00 yields 17 rows).
- **Segment Sub-divisions:** Each hour row is sliced horizontally into a user-defined number of equal sub-segments (1 to 6 segments per hour).
- **Segment Internal Layout (Always Left-Aligned):**
 - **Time Indicator:** A passive, grayed-out, 2-digit identifier. The first segment of an hour always displays the hour (e.g., 07); subsequent segments display their starting minute (e.g., 15 , 30 , 45). This utilizes the secondary monospace font.
 - **Interactive Goal Text Field:** A borderless, flat, inline text input box.
 - **Overflow Handling:** If a goal text exceeds the available horizontal space of the segment column, it is truncated gracefully using a clean ellipsis (. . .).
- **Contextual State Boundaries:**
 - **Future & Present Segments:** Goal text remains completely interactive and editable.
 - **Past Segments:** Goal text inputs are automatically locked, grayed out, and uneditable.
 - **Completion Validity:** Checking/Unchecking goals can only be performed on the current, active segment (via Focus Mode or the main visualization controls).

Slide 3: Settings & Customization (Bottom Screen)

Provides comprehensive structural, typography, color, and file manipulation settings.

- **Structural Configuration:** Controls to customize the active starting hour, ending hour, and segments per hour (1–6).
- **Design Token Configuration:**
 - **Relative Font Scaling:** A slider to set the height of typography relative to the segment height (e.g., default 50%, adjustable from 25% to 75%).

- **Color Swatches:** Direct customization inputs for application background, regular text, highlight text, interactive elements, success, failure, active progress, unreached future background, grid lines, and grid stroke thickness.
- **Functional Behavior Switches:**
 - **Status Coloring Toggle:** On/off switch to determine if missed segments turn to the "Failure" color, or if they simply keep the standard neutral past-progress bar color.
- **Data Portability Engine:**
 - **Export to CSV:** Serializes all system parameters, design tokens, and daily goals into a flat, well-formatted CSV file. This enables users to easily modify their agenda in Google Sheets or Excel and re-import.
 - **Import from CSV:** Accepts file uploads and parses them using standard rules. It checks constraints (valid times, segment ranges, and string lengths) before updating the state to prevent application breakage.
- **Native System Sharing:** A clean button invoking the browser's native sharing overlay. It shares only the bare application URL to protect user privacy.
- **Credits:** An ultra-minimal, unembellished footnote line.

4. Enhanced Interaction & Utility Features

A. Dynamic Tab Title & Favicon Tracking

- **Tab Title Format:** Displays the live countdown and goal details: `[MM:SS] | Goal Title - howfar.today`
 - The timer matches the remaining minutes and seconds of the active segment and updates every second.
 - If no goal is configured for the active segment, the display falls back to: `[MM:SS] | Focus - howfar.today`
- **Dynamic SVG Favicon:** The page header's favicon updates in real-time. It renders a clean, inline SVG data URL that dynamically alters color/state (e.g., a simple colored block or ticking circle representing current progress and status: success, failure, or neutral).

B. Fluid Keyboard Grid Navigation

To maximize efficiency and eliminate the need for mouse interaction during rapid setup or editing, the grid supports a native keyboard navigation layer:

- **Enter:** Saves the active goal input, focuses immediately on the next chronological segment's input field, and automatically selects its text for overwriting/editing.
- **Shift + Enter:** Saves the active goal input and focuses on the immediately preceding segment's input field.
- **Escape:** Saves the current input value, removes focus/blurs the field, and restores global page-level viewport slide-scrolling (Up/Down arrow navigation).

5. Design Tokens Matrix

--bg-app:	[App background canvas]
--color-regular:	[Standard text display]
--color-highlight:	[Focused/active states]
--color-interact:	[Clickable items/inputs]
--color-secondary:	[Muted time indicators and secondary details]
--color-success:	[Completed segment progress background]
--color-failure:	[Failed segment progress background]
--color-progress:	[Active running progress track fill]
--color-void:	[Unreached future segment background]
--color-grid:	[Grid structural dividing lines]
--stroke-grid:	[Grid line thickness (relative unit)]

Primary Font Color **Secondary Font Color** **Grid Line Color** **Progress Track Color** **Failure Background Color**